IDS 705 Final Report

Vulnerability of Medical AI: Evaluating Adversarial Attacks on Diagnostic Imaging

A study on the robustness of Convolutional Neural Networks using MedMNIST

Team Members:

Shelly Cao
Arvind Kandala
Diwas Puri
Sebine Scaria

Team Identifier:

Group 10

Target stakeholder(s) for this report: Hospitals, Insurance company, Hospital IT Security Directors, Medical AI Compliance Officers, and Healthcare Software Vendor.

Executive Summary

Hospital-deployed AI tools for pneumonia detection are increasingly trusted to inform clinical decisions, yet a critical question remains unanswered: do these models hold up when real-world imaging conditions are imperfect? Chest X-ray images vary in brightness, contrast, and sharpness across different hospitals and scanners, variations that were absent during model training. If performance collapses under these conditions, patients with pneumonia may be incorrectly cleared as healthy, creating a direct patient safety risk.

We stress-tested a pneumonia detection AI by deliberately degrading chest X-ray image quality in ten different ways and measuring how much diagnostic accuracy suffered. Our key finding is that certain quality issues, particularly low contrast and abnormal brightness, caused the model to stop distinguishing between sick and healthy patients entirely. A targeted manipulation attack caused the opposite problem: the model began confidently clearing pneumonia patients as healthy while flagging normal ones. In both failure scenarios, the model gave no warning that anything was wrong, leaving clinicians with no signal to know when to distrust it. AI diagnostic tools that appear highly accurate under ideal conditions can become unreliable or dangerous in real hospital environments, and this risk is currently invisible to those relying on them

Decisions to be made:

Hospital IT Security Directors should consider adding safeguards to check image quality before it reaches AI systems, since routine issues (e.g., aging scanners or data transfer) can cause silent failures.
Medical AI Compliance Officers should reassess approval standards: testing on clean images isn't enough. Models must be evaluated under real-world degradation, and systems vulnerable to targeted attacks may require stricter review due to higher risk.
Healthcare Software Vendors should build AI that can say "I'm not sure" when image quality is poor, instead of giving confident but wrong predictions.

Motivation

Medical AI systems for pneumonia detection have achieved clinical-grade accuracy on benchmark datasets, yet their robustness under real-world imaging conditions remains poorly understood. In practice, chest X-rays acquired across different hospitals, scanner manufacturers, and imaging protocols vary in brightness, contrast, resolution, and sharpness. Equipment aging, suboptimal exposure settings, and lossy image compression during transmission can all introduce systematic distortions that were absent during model training. If diagnostic performance degrades significantly under such conditions, deployment in clinical settings poses a direct patient safety risk: a patient with pneumonia may be incorrectly classified as healthy, with no warning signal to the clinician reviewing the result.

Report

Introduction

Medical imaging AI has moved rapidly from research benchmark to clinical tool. Radiology departments, telemedicine platforms, and diagnostic software vendors increasingly rely on deep learning models to flag abnormalities in chest X-rays. The images these models were trained on, however, rarely match the images they encounter in practice.

A chest X-ray does not travel cleanly from acquisition to inference. It passes through storage systems, gets compressed for transmission, is resized by preprocessing pipelines, and may be displayed on monitors with inconsistent calibration. Each step can alter the pixel values the model reads, sometimes in ways invisible to a human reviewer but meaningful to an algorithm trained on clean benchmark data.This report evaluates how much that gap matters.

Using models trained on pediatric pneumonia X-rays, we tested ten forms of image corruption and adversarial manipulation across five severity levels to understand where model reliability breaks down and what that means for patients on the other end of a missed diagnosis. We evaluated three model architectures — ResNet-18, ResNet-50, and our own lightweight custom CNN, DASYNet — under identical conditions to determine whether vulnerability is model-specific or systemic.

Data

We tested our model on the PneumoniaMNIST+ benchmark, consisting of 5,856 pediatric chest X-rays. On clean, unmodified images, ResNet-18 achieves an 89.42% accuracy and a 97.44% recall, successfully identifying 380 of 390 pneumonia cases. This baseline represents our performance ceiling; any subsequent drop in accuracy or recall caused by image corruption directly erodes the safety margins essential for reliable clinical diagnostics.

What We Tested

We applied ten types of image corruption across five severity levels to simulate what happens to X-rays in real clinical pipelines. Nine attacks are organized into three categories: structural attacks (Gaussian Blur, Downsampling, JPEG compression), photometric dark attacks (Brightness Dark, Contrast Low, Gamma Underexposed), and photometric bright attacks (Brightness Bright, Contrast High, Gamma Overexposed). The tenth is FGSM, an adversarial attack that makes invisible but mathematically precise edits to fool the model. Each attack is parameterized from L1, a slight perturbation, to L5, an extreme one.

Figure 1 shows two corruption types across severity levels: Gamma Underexposed and FGSM. Gamma underexposure progressively darkens the image, suppressing the contrast between healthy lung tissue and areas of consolidation that the model relies on to detect pneumonia. FGSM looks nearly identical to the original at low severity but is mathematically engineered to exploit the model's blind spots. By Level 3, both corruptions are doing meaningful damage to model performance while still appearing as plausible clinical images to a human reviewer.

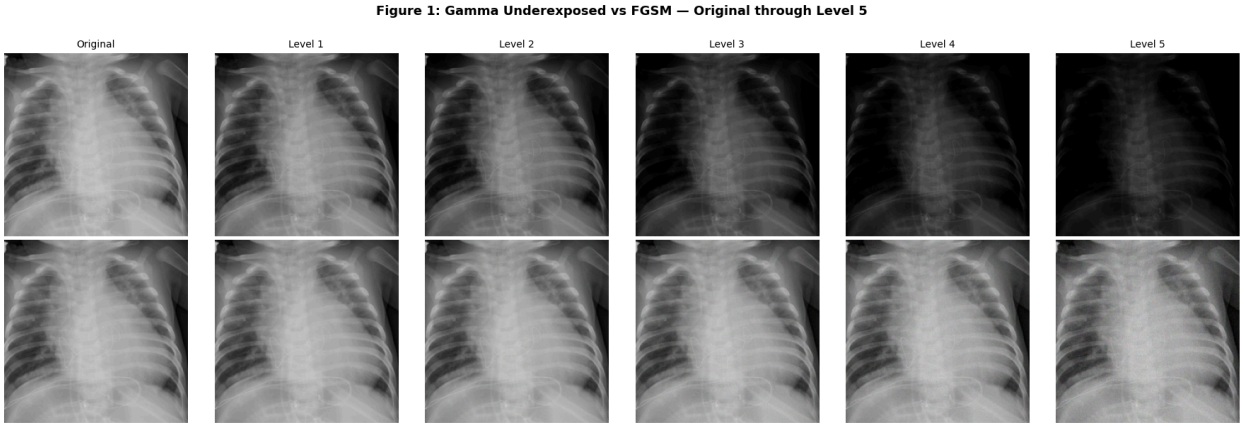


Figure 1. Two representative corruption types across five severity levels. Gamma Underexposed progressively darkens the image, eroding the contrast between lung tissue and areas of consolidation. FGSM introduces perturbations that are visually indistinguishable from the original at low severity yet are precision engineered to mislead the model. Level 3 represents the realistic upper bound of what a deployed system might silently encounter without triggering a quality flag.

To ensure our findings were not specific to a single architecture, we evaluated two distinct models. While ResNet-18 served as our deep-learning baseline, we also developed and tested DASYNet, a custom-engineered CNN tailored for 224x224 grayscale medical images.

DASYNet utilizes a streamlined architecture of three convolutional blocks and AdaptiveAvgPool2d, resulting in a lightweight footprint of only 97,634 parameters. This makes it an ideal candidate for resource-constrained clinical environments, though it presents unique robustness trade-offs compared to the more complex ResNet baseline.

Key Findings

The results tell a clear and consistent story across all three analyses. Not every corruption is equally dangerous, not every severity level represents a realistic threat, and not every part of the image carries equal weight. What follows highlights the findings most relevant to deployment decisions: which attack types cross the threshold into clinical risk, at what point they do so, and where in the image the model is most exposed. By looking at Figure 1, we chose Level 3 severity attack, since it passes the human eyes. It is also the optimal attack level to test to see how well the three models work.

Figure 2 tracks how accuracy and recall change as each corruption intensifies. Gaussian Blur is included as a reference point: performance holds steady through all five levels, illustrating what a resilient corruption looks like and giving the other curves something meaningful to be compared against.

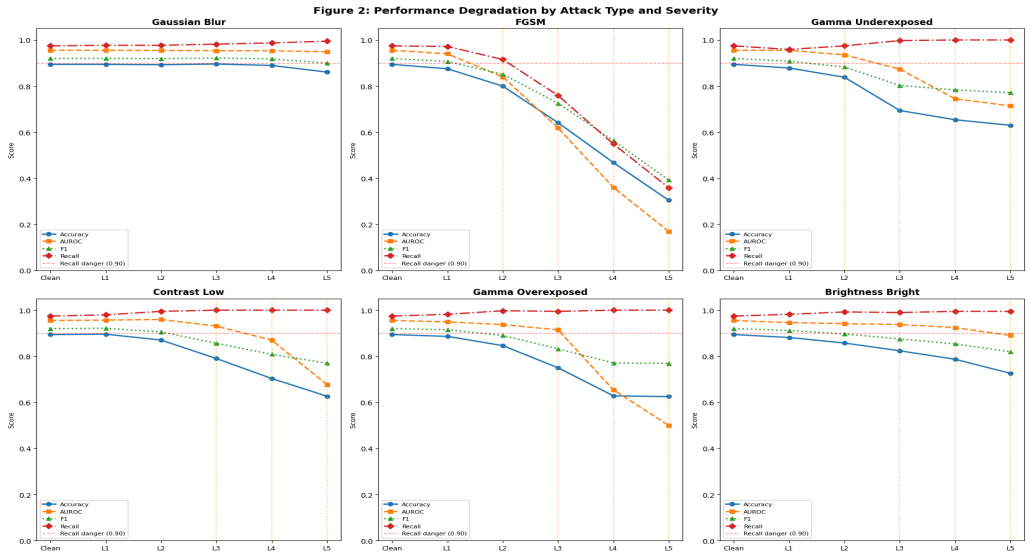
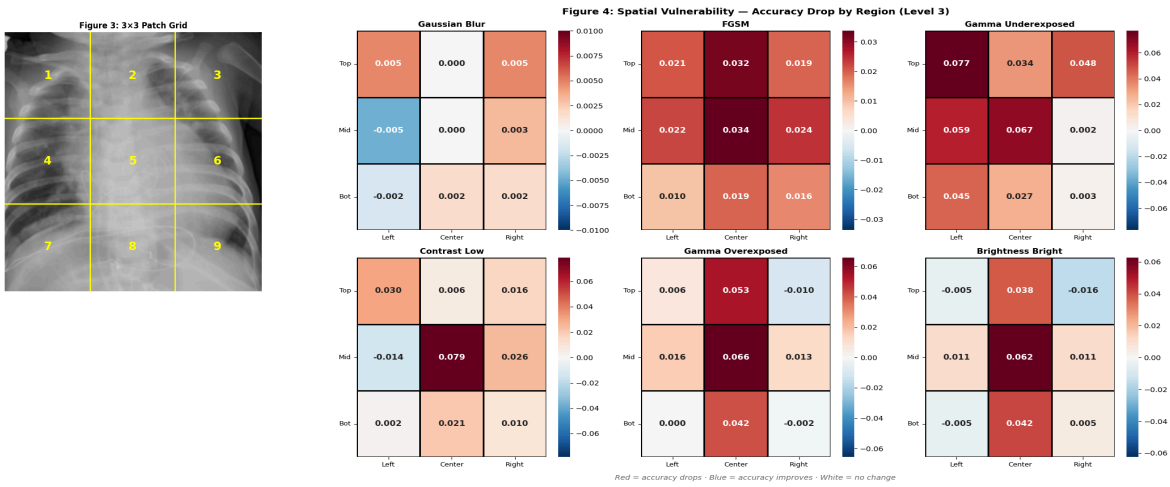


Figure 2: Performance metrics for ResNet-18 under six corruption types. The blue line represents accuracy, while the solid red line shows recall, with a dashed red line marking the 0.90 clinical safety threshold. Orange vertical markers indicate severity levels where accuracy drops >5%. While Gaussian Blur serves as a resilient baseline, FGSM triggers immediate performance collapse. Conversely, Gamma and contrast degradations show accuracy erosion while recall remains high, indicating the model is defaulting to flagging nearly all cases as pneumonia.

FGSM tells the opposite story. Even at the reduced epsilon range used here, all metrics decline sharply from Level 1 onward, reaching 30% accuracy and 36% recall by Level 3, making it the only attack that produces a genuine and measurable increase in missed pneumonia diagnoses. The gamma and contrast attacks present a different but equally serious failure mode: accuracy collapses to around 63% while recall climbs toward 100%, meaning the model has stopped discriminating altogether and is flagging nearly every image as pneumonia. Both outcomes are clinically unacceptable, just in opposite directions.



Figures 3–4. Patch-Grid Spatial Vulnerability Analysis — ResNet-18 at Level 3 Severity

Each chest X-ray is divided into nine equal regions using a 3×3 grid (Figure 3). Corruptions are applied to one region at a time while the rest of the image remains clean, isolating the contribution of each spatial location to model vulnerability. Figure 4 shows the resulting accuracy drop per region at Level 3 for the five most damaging attacks alongside Gaussian Blur as a low-impact reference. Red cells indicate regions where localized corruption most severely hurt model performance; blue cells indicate regions where corruption had negligible or no effect. The center square is consistently the most vulnerable position across all attack types, reflecting the anatomical importance of the mediastinum and hilar regions where pneumonia consolidation most commonly appears. Gaussian Blur produces negligible change regardless of region, confirming that spatial location matters far more for attacks that exploit the model's learned features than for those that simply reduce image sharpness.

Our experiments comparing ResNet-18 and ResNet-50 revealed that, despite similar baseline performance, ResNet-50 is paradoxically more vulnerable to environmental corruption. While ResNet-18 showed notable resilience to severe perturbations—such as Gaussian blur, where accuracy dropped by only 3.4%—ResNet-50 exhibited significantly higher sensitivity, with performance degradation more than doubling in similar scenarios.

This trend persisted across patch-based stress tests, though we observed distinct spatial vulnerabilities between the models. While the center patch remained the most critical point of failure for both, their reactions to edge-case degradations varied, suggesting that model architecture influences which specific spatial features are most susceptible to noise. Given that ResNet-50 contains over double the parameters of ResNet-18 (≈ 23.5 million vs ≈ 11.2 million), this increased susceptibility likely stems from overfitting to our limited training set of 4,708 images.

Table 1: Attacked metrics reflect performance under Gamma Overexposed corruptions at Severity Level 3. FGSM Drop reports the accuracy decline at Level 1 (ResNet-18) and Level 1 (ResNet-50) relative to each model's clean baseline.

	ResNet-18		**ResNet-50**	
Metric	Clean	Attacked*	Clean	Attacked*
Accuracy	89.42%	75.00%	88.94%	71.31%
Recall	97.44%	99.49%	97.69%	99.49%
AUROC	0.9553	0.9145	0.9698	0.8904
FGSM Drop (L1)	–	−1.92%	–	−61.86%

To confirm these vulnerabilities are not model-specific, we replicated the study on DASYNet, a custom lightweight CNN built for medical imaging. Despite strong clean-data performance, DASYNet proved even more fragile than ResNet-18 under stress. Digital interference caused near-total diagnostic collapse, and the same high-risk lighting and contrast vulnerabilities emerged — particularly overexposure, which produced severe performance drops regardless of model architecture. Spatial analysis further revealed a consistent failure point: corruptions concentrated in the central scan region, where the lungs and heart appear, reliably triggered diagnostic breakdown. Together, these findings suggest the vulnerabilities identified in this study are architectural in nature, not artifacts of any single model.

To test whether training on corrupted data improves robustness, we fine-tuned DASYNet on a corruption-augmented dataset combining clean PneumoniaMNIST+ images with corrupted

counterparts across all five attack types, producing 31,392 total training images. The model was fine-tuned for 5 epochs at a learning rate of 1e-4 and evaluated on the official validation split under both clean and corrupted conditions.

Table 2: DASYNet performance before and after corruption-aware fine-tuning at Level 3 severity. Accuracy, Recall, and AUROC are reported for each of the five selected attacks and the clean baseline.

Attack	**Accuracy**		**Recall**		**AUROC**	
	Pre-FT	Post-FT	Pre-FT	Post-FT	Pre-FT	Post-FT
Clean (baseline)	0.9006	0.8974	0.9744	0.9872	–	–
Gamma Underexposed	0.7212	0.8718	0.9949	0.9769	0.8923	0.9558
Contrast Low	0.6282	0.8846	1.0000	0.9923	0.6501	0.9655
Gamma Overexposed	0.7756	0.8638	1.0000	0.9949	0.9472	0.9670
Brightness Bright	0.8990	0.9087	0.9744	0.9872	0.9614	0.9661

Note: For reference, ResNet-18 achieves a clean accuracy of 89.42% on the same test

Takeaway

Our cross-architectural analysis proves that model complexity is not a surrogate for clinical safety. Despite significant differences in scale—ranging from the lightweight DASYNet to the deep ResNet-50—all models exhibited identical failure modes when subjected to environmental and adversarial stress. This reveals a systemic vulnerability in medical AI to common variations like gamma overexposure and low contrast, which are frequently encountered in diverse hospital hardware environments. For leadership, the primary takeaway is that high performance on clean training data does not guarantee reliability in a live clinical setting. To address these risks, we recommend that healthcare institutions transition from static accuracy benchmarks to rigorous operational robustness testing. Our findings demonstrate that while diagnostic integrity can collapse by over twenty-five percent under common lighting shifts, these failures are preventable through corruption-aware fine-tuning and standardized preprocessing like histogram equalization. Hospitals should mandate that vendors provide robustness certificates for various image conditions. Furthermore, the catastrophic failure of all models under adversarial stress underscores the need for end-to-end encryption of the imaging pipeline to prevent targeted data manipulation and protect patient safety.

Limitation

This study has three primary limitations. First, all corruptions are applied digitally in simulation: we did not evaluate models on images acquired under genuinely degraded hardware conditions, which may produce different artifact structures than our parameterized transforms. Second, the PneumoniaMNIST+ dataset is derived from a single pediatric institution, and vulnerability characteristics may differ across adult populations, different demographic groups, or different scanner types. Third, DASYNet's extreme sensitivity to adversarial noise, which produced an 88.3% accuracy drop under FGSM at Level 5, is likely compounded by its limited model capacity, and findings for DASYNet should not be generalized to all lightweight architectures. Actual clinical performance may also vary when using high-resolution native X-ray formats or when multiple environmental stressors occur simultaneously.

References:

A. Madry, A. Makelov, L. Schmidt, D. Tsipras, and A. Vladu, "Towards Deep Learning Models Resistant to Adversarial Attacks," *International Conference on Learning Representations (ICLR)*, 2018.

Finlayson, S. G., Bowers, J. D., Ito, J., Zittrain, J. L., Beam, A. L., & Kohane, I. S. (2019). Adversarial attacks on medical machine learning. Science, 363(6433), 1287–1289. https://doi.org/10.1126/science.aaw4399

I. Goodfellow, J. Shlens, and C. Szegedy, "Explaining and Harnessing Adversarial Examples," *International Conference on Learning Representations (ICLR)*, 2015.

Appendix A: Background and Related Work

Around 2019, researchers begin to realize that the adversarial vulnerabilities in medical machine learning can pose a serious threat to the well-being of patients as doctors may begin to place high reliability on these models. People including patients, providers, and insurers may have financial incentives to manipulate AI outputs, and this concern should bring in engagement from medical, legal, and ethical communities (Finlayson, 2019). This brings about the question of what types of attacks pose the most concern for this case. Beyond deliberate adversarial attacks, medical machine learning systems may encounter images that have natural variation/noise stemming from imaging equipment and compression during transmission. Things like shifts in brightness, contrast, resolution, and sharpness can create a shift in the underlying probability distribution that characterizes the images, which can degrade model performance without any malicious intent. Our study evaluates these corruptions organized into structural attacks (Gaussian blur, downsampling, JPEG compression) and photometric attacks (brightness and contrast shifts in both directions, gamma correction), . These categories were chosen to reflect the specific failure modes most likely to occur in real clinical imaging pipelines.

The specific attacks that we evaluate are based on prior work on adversarial robustness. Goodfellow et al. (2015) created the Fast Gradient Sign Method (FGSM), which corrupts the image by taking a step in the direction of the gradient of the loss with respect to the input image. FGSM only requires one forward and backward pass through the model to implement, which makes it a computationally cheap attack to implement for an adversarial user who manages to gain access to these images. We chose to use the residual network (ResNet) models because they allow for shortcut connections between layers, which in turn allows gradients to be preserved (reduces vanishing gradients) and enables good performance on image classification tasks. Comparing the robustness of these two models under identical corruption conditions allows us to ask whether the added depth and parameter count of ResNet-50 actually increases performance ability.

Appendix B: Methods and Experimental Design

Dataset, Models, and Preprocessing

We used the PneumoniaMNIST test split from the MedMNIST benchmark, which consisted of 624 images normalized to mean 0.5 and standard deviation 0.5 across all channels. MedMNIST trained a ResNet-18 and ResNet-50 on PneumoniaMNIST and published the weights on their Zenodo repository. We loaded these weights without any fine-tuning, so all ResNet experiments are purely evaluative.

Attack Design and Evaluation Protocol

The nine clinically motivated corruptions were implemented as deterministic, parameterized transforms applied at inference time. Gaussian Blur, Downsampling, and JPEG compression were applied before tensor conversion, while Brightness Dark/Bright, Contrast Low/High, and Gamma Overexposed/Underexposed were applied after normalization. Each attack was divided into five monotonically spaced severity levels such that Level 5 is extremely degraded and Level 1 is barely perceptible to the human eye. Hence, we chose Level 3 since it not only passes the human eye, but is also a reasonable level of attack. Metrics were computed using the official MedMNIST evaluator for AUC and accuracy, and scikit-learn for precision, recall, and F1. The classification threshold was fixed at 0.5. Because avoiding false negatives is especially critical in the context of a potentially life-threatening disease, we designated a recall danger threshold of 0.90, below which a model was considered clinically unsafe regardless of its overall accuracy. A secondary robustness verdict was derived from the accuracy drop relative to the clean baseline: drops below 2% were rated Robust, 2–5% Concerning, and above 5% Vulnerable. Combined attacks were evaluated on all valid pairs from the nine corruption types, excluding three conflict pairs that are physically contradictory: Brightness Dark with Brightness Bright, Contrast Low with Contrast High, and Gamma Overexposed with Gamma Underexposed.

Because the spatial distribution of diagnostically relevant features across a chest X-ray is highly non-homogenous, we hypothesized that the same corruption applied to different image regions would produce meaningfully different effects on model predictions. The lung parenchyma and hilar regions occupy the central portion of the image, while the peripheral and outer regions capture anatomically less informative content. To test this, each of the 10 attack types was applied in isolation to each of nine equal 3×3 grid squares at Levels 1, 2, and 3, while the rest of the image remained clean. For each of the 270 combinations (10 attacks × 9 regions × 3 severity levels), inference was run over the full 624-image test set and accuracy drop relative to the clean baseline was recorded. This isolates spatial sensitivity from overall attack severity.

DASYNet Architecture and Training

We also developed a custom CNN model called DASYNet. The DASYNet architecture is a lightweight convolutional neural network designed to process 224×224 grayscale medical images while maintaining a low parameter count suitable for deployment in resource-constrained clinical settings. Unlike the ResNet-18 baseline, which contains approximately 11 million parameters, DASYNet consists of 97,634 parameters, enabling faster inference at the cost of reduced representational capacity.

The architecture consists of three sequential convolutional blocks, each comprising a 3×3 convolution followed by batch normalization, ReLU activation, and max-pooling to progressively reduce spatial dimensions while retaining salient diagnostic features. An adaptive average pooling layer then collapses the feature maps into a fixed 3×3 representation, which is passed through a fully connected classifier for binary output. This design ensures that the model accepts inputs of any spatial resolution, making it compatible with the 224×224 MedMNIST images used throughout this study.

For initial training, DASYNet was optimized using AdamW with a learning rate of 1e-3, a weight decay of 1e-4, and a batch size of 192. Training ran for up to 35 epochs with early stopping based on validation loss, using a patience of 8 epochs, and the best checkpoint by validation loss was retained. For the corruption-aware fine-tuning described in the defensive strategy section, a lower learning rate of 1e-4 was used to preserve the features learned during initial training. The fine-tuning dataset combined clean PneumoniaMNIST train and validation images with corrupted versions generated from four of the five selected attack types at Level 3 severity. FGSM was excluded from the fine-tuning mix because it requires gradient computation and cannot be applied as a standard data augmentation transform. Detailed layer specifications and model weights are preserved in the associated project repository for reproducibility.

Corruption-Aware Fine-Tuning

To evaluate whether exposure to corrupted inputs during training improves model robustness at inference time, we conducted a corruption-aware fine-tuning experiment using DASYNet as the target architecture. DASYNet is a compact CNN designed to operate directly on grayscale 224×224 inputs. It processes images through three convolutional blocks, each consisting of a 3×3 convolution, batch normalization, ReLU activation, and 2×2 max pooling, progressively expanding the channel dimension from 16 to 64. A global adaptive average pooling layer reduces spatial dimensions to 3×3, and a two-layer classifier with 128 hidden units and 0.5 dropout produces the final binary prediction.

For fine-tuning, we constructed a combined dataset by pooling the clean PneumoniaMNIST+ training split (4,708 images), the clean validation split (524 images), and the corrupted train and validation counterparts across all five attack types (Gamma Underexposed, Gamma Overexposed, FGSM, Contrast Low, and Brightness Bright), each applied at Level 3 severity. This yielded 31,392 total images, of which 8,094 were labeled normal and 23,298 labeled pneumonia, preserving the class imbalance present in the original dataset. All images were resized to 224×224, converted to single-channel grayscale, and normalized to zero mean and unit variance using a mean and standard deviation of 0.5.

Fine-tuning proceeded for 5 epochs with a batch size of 64 and a learning rate of 1e-4 using the Adam optimizer and cross-entropy loss. Model performance was evaluated on the official PneumoniaMNIST+ validation split under six conditions: clean images and each of the five corruption types individually, as well as the combined corrupted validation pool. This setup allowed a direct comparison between the robustness of the fine-tuned DASYNet and the frozen pretrained ResNet-18 under identical attack conditions.

Appendix C: Ethical Considerations

This project occupies a dual-use research space: by demonstrating how medical AI systems can be made to fail, we create knowledge that could theoretically be used to harm patients through deliberate manipulation of diagnostic systems. We address this concern through several design choices. All experiments were conducted on a standardized public benchmark dataset (PneumoniaMNIST) with no involvement of real patient data from active clinical systems. The attacks we evaluate are primarily clinically realistic corruptions that occur naturally in imaging pipelines, not sophisticated adversarial perturbations requiring insider access. Documenting these vulnerabilities is a prerequisite for mandating the robustness standards that will protect patients.

Regarding potential sources of bias: PneumoniaMNIST is derived from a pediatric population at a single institution (Guangzhou Women and Children Medical Center). The baseline model's performance and vulnerability characteristics may differ for adult populations, different demographic groups, or different imaging equipment. Our vulnerability findings should therefore be interpreted as evidence for a specific model and dataset, not as universal claims about all pneumonia detection systems. We explicitly flag this limitation.

Privacy considerations are minimal: we used a fully anonymized, publicly available dataset distributed under the Creative Commons Attribution 4.0 license, with all identifying information removed prior to release by the original dataset authors. No new patient data was collected or accessed.

The most significant potential negative consequence of this work is that the documented vulnerability to contrast and gamma attacks could be exploited by malicious actors to systematically cause misdiagnosis. We consider this risk to be outweighed by the benefit of disclosure: these vulnerabilities exist regardless of whether they are documented, defensive countermeasures can only be developed once vulnerabilities are characterized, and the attacks we document are discoverable by any competent adversary. Responsible disclosure of AI system vulnerabilities is increasingly recognized as a necessary step in the development of trustworthy medical AI.

Transparency and explainability are central to the value of this work. The Grad-CAM visualizations produced alongside this analysis provide clinicians with insight into model decision-making, supporting human oversight of AI recommendations. We recommend that any clinical deployment of diagnostic AI include explainability interfaces that allow radiologists to understand and override model decisions, particularly given our finding that this model provides no confidence signal to indicate when it has entered a catastrophic failure state.

Appendix D: Individual Roles

Shelly Cao

Shelly led the data acquisition, preprocessing pipeline and exploratory data analysis. She established the standardized data loading pipeline using MedMNIST+ at native 224x224 resolution, implemented the Gaussian blur attack pipeline, contributed to the patch-grid visualization analysis, and led the compilation of the final report.

Arvind Kandala

Arvind implemented the baseline ResNet-18 model pipeline and established the evaluation framework using the official MedMNIST Evaluator. He implemented the JPEG compression attack, verified clean baseline performance against published benchmark results, and designed the robustness verdict thresholds used throughout the evaluation.

Diwas Puri

Diwas engineered the FGSM adversarial attack pipeline and implemented the downsampling attack. He ran the primary single-attack and pairwise combined-attack experiments, produced the attack ranking tables, and performed the worst-case analysis identifying Gamma Overexposed and Contrast Low as the most dangerous attack types.

Sebine Scaria

Sebine implemented the brightness and contrast attack pipelines, gamma correction attacks, and the 3x3 patch-grid regional vulnerability analysis with full metric sweeps across all severity levels. She implemented the combined brightness and contrast attack analysis, performed the model confidence analysis revealing 100% confidence under catastrophic failure, implemented Grad-CAM explainability visualizations to show attention shift under attack, and developed the histogram equalization defense framework as a baseline defense strategy.